实验六.端到端车道保持

实验目的

1. 了解车辆自动驾驶端到端的概念

2. 熟悉 AI 模型数据采集、标注、训练、转换和部署的过程

实验过程

1.数据采集

在/userdata/pictures/目录下，新建 end2end/image_dataset 文件夹；

上传 picker_end.py 到/userdata/目录下，打开该文件，修改下面两处地方：

```
def cmd_vel_callback(self, msg):
    # 记录每一次收到的指令＆当前采集到的画面
    self.num +=1
    if self.num ==10:
        img_name = 'zx %f %f %s' % (msg.angular.z,msg.linear.x, uuid.uuid1())
        self.image = self.image[100:548, :, :]    #取车道线兴趣区域
        cv2.imwrite('/userdata/picture/end2end/image_dataset/' + img_name + '.jpg',
        self.num =0
```

self.num==10，代表 10 抽 1，采样数据可根据需要，自行修改；

self.image[100;548,:,:]，代表 h1 是 100，h2 是 548，车道线兴趣区域。根据实验 5 修改 h1 和 h2。

在第一个终端里，启动机器人底盘驱动：

ros2 launch originbot_bringup originbot.launch.py use_camera:=true

第二个终端，运行手柄控制节点：

ros2 launch originbot_bringup joy_teleop.launch.py

在第三个终端，采集数据，输入：

cd /userdata

python picker_end.py

用手柄遥控小车沿车道线行驶，观察 image_dataset/目录下，是否有数据集

产生。

建议顺时针和逆时针各跑两圈后，结束采样。下载 image_dataset 文件夹到当地硬盘。

附录 picker_end.py

```
#!/usr/bin/env python
# -*- coding: utf-8 -*-

import rclpy, cv2, cv_bridge
from rclpy.node import Node
from sensor_msgs.msg import CompressedImage
from geometry_msgs.msg import Twist
import uuid
import os

class Picker(Node):
    def __init__(self):
        super().__init__('end2end_image_recorder')
        self.get_logger().info("start picking!")
        self.bridge = cv_bridge.CvBridge()

        self.image_sub = self.create_subscription(CompressedImage, '/image_jepg', self.image_callback, 10)  self.cmd_vel_sub = self.create_subscription(Twist, '/cmd_vel', self.cmd_vel_callback, 10)
        self.num=0

    def image_callback(self, msg):
        self.image = self.bridge.compressed_imgmsg_to_cv2(msg)

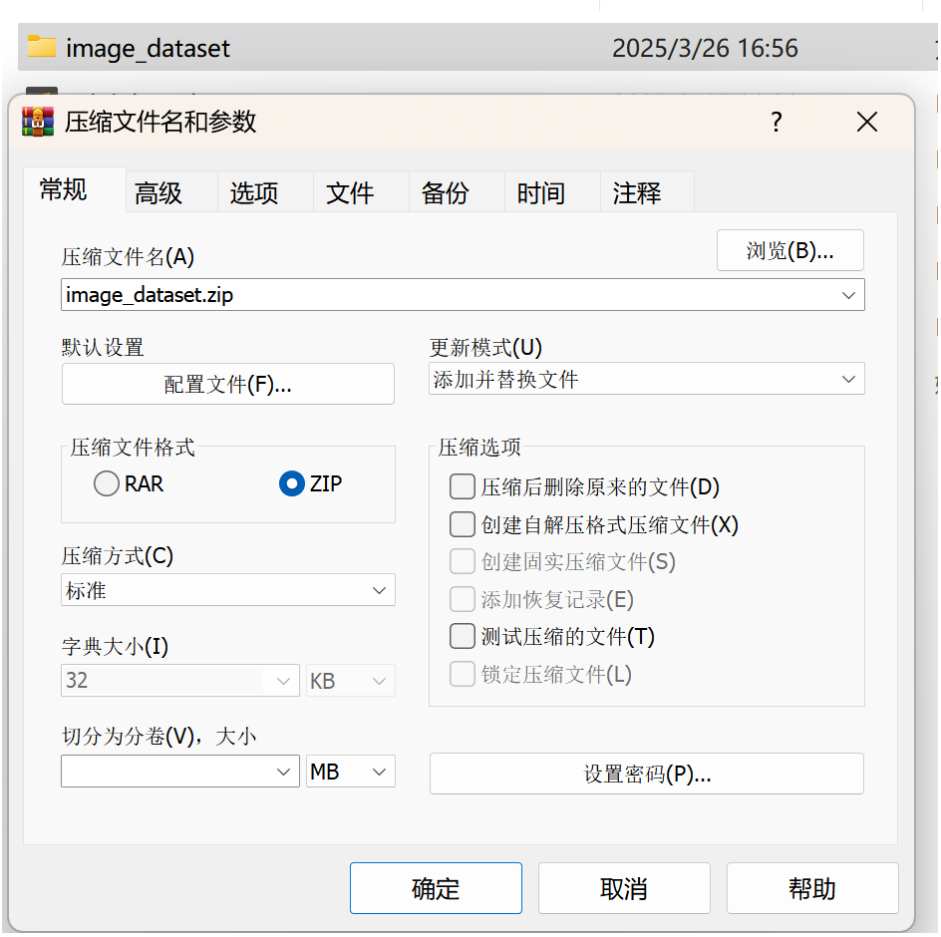
    def cmd_vel_callback(self, msg):
        # 记录每一次收到的指令＆当前采集到的画面
        self.num +=1
        if self.num ==10:
            img_name = 'zx_%f_%f_%s' % (msg.angular.z,msg.linear.x, uuid.uuid1())
            self.image = self.image[100:548, :, :]    #取车道线兴趣区域
            cv2.imwrite('/userdata/pictures/end2end/image_dataset/' + img_name + '.jpg', self.image)
            self.num =0

def main(args=None):
```

```
    rclpy.init(args=args)
    picker = Picker()
    try:
        rclpy.spin(picker)
    except KeyboardInterrupt:
        pass
    finally:
        picker.destroy_node()
        rclpy.shutdown()
if __name__ == '__main__':
    main()
```

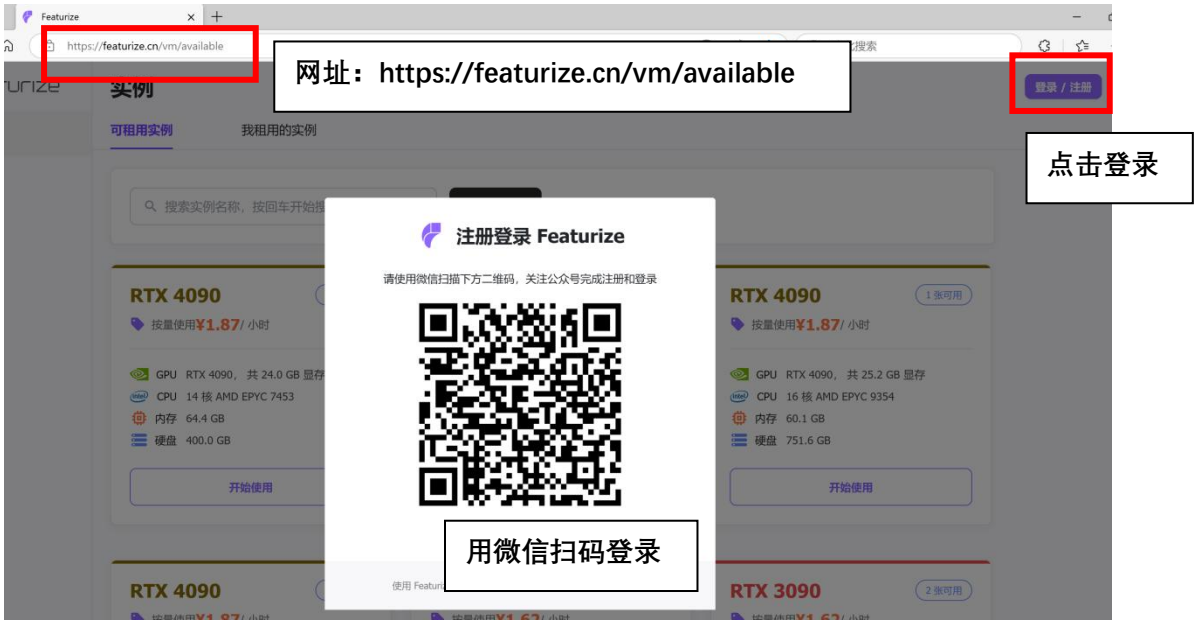
2、训练模型（云资源）

把数据集 image_dataset 压缩成一个 zip 文件。



登陆网站 https://featurize.cn/vm/available

点击登录/注册，用微信扫二维码

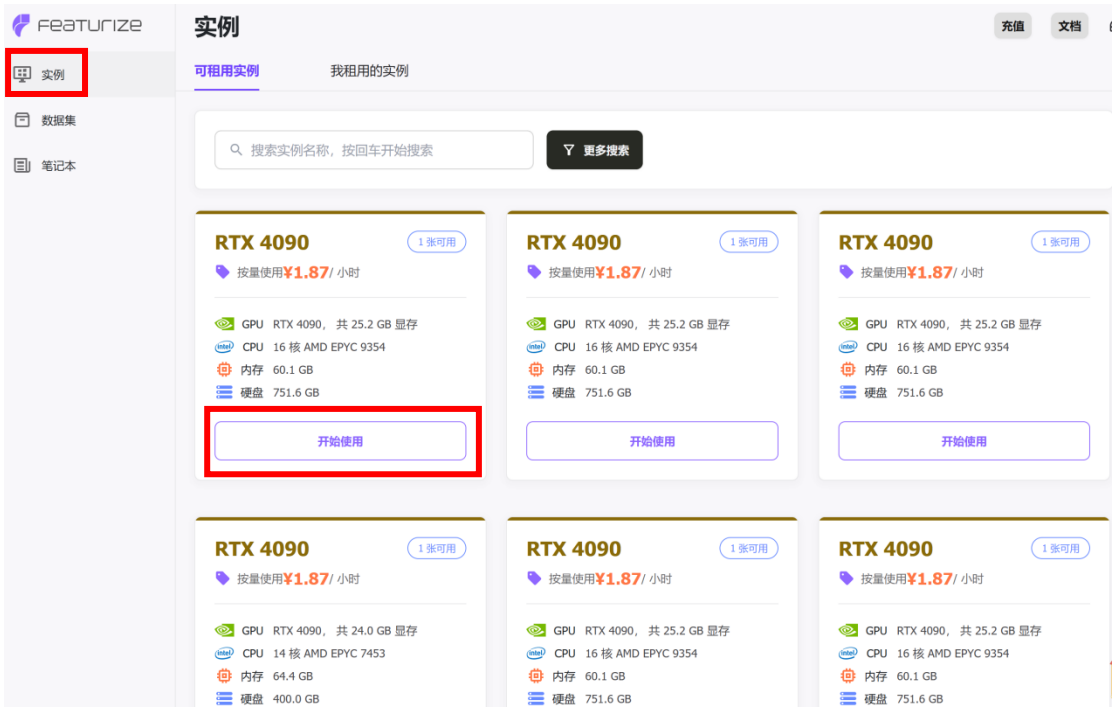


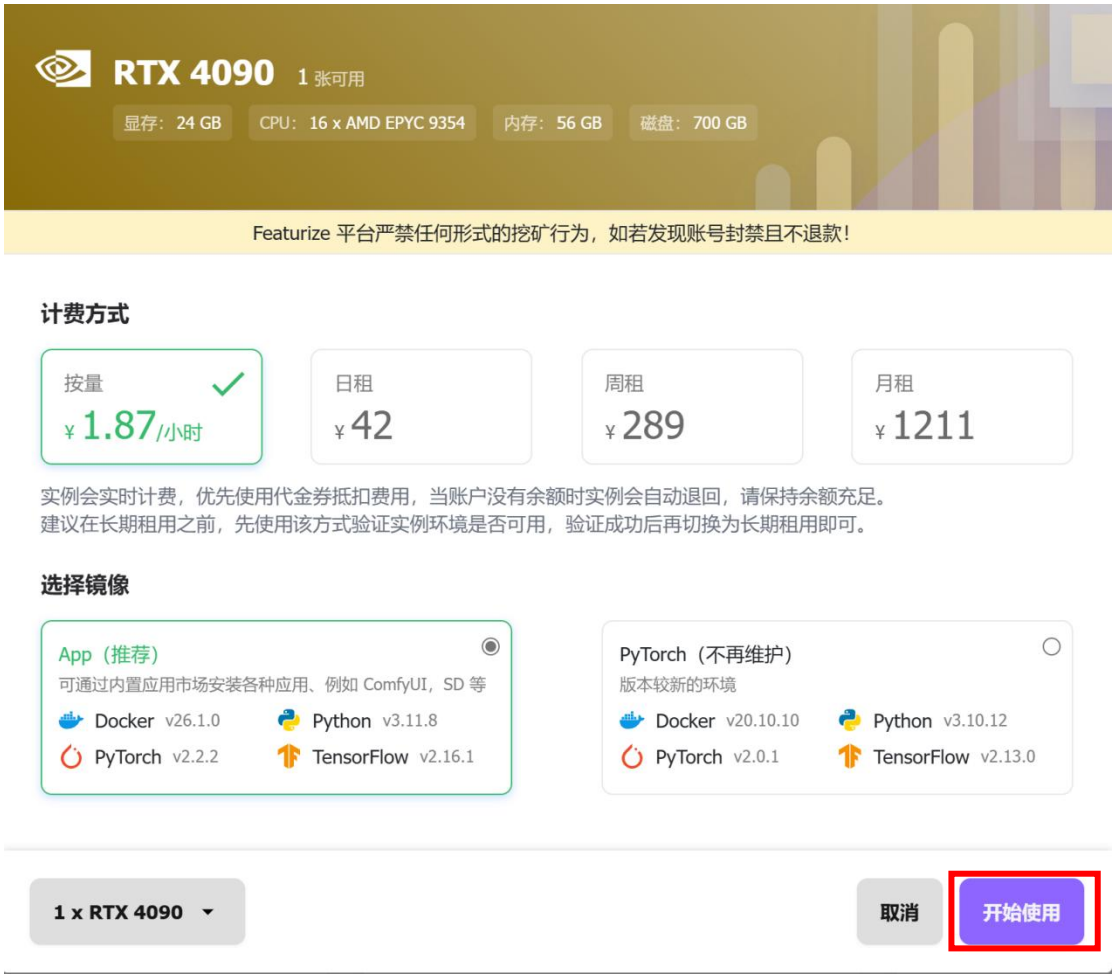
点击充值，输几块钱



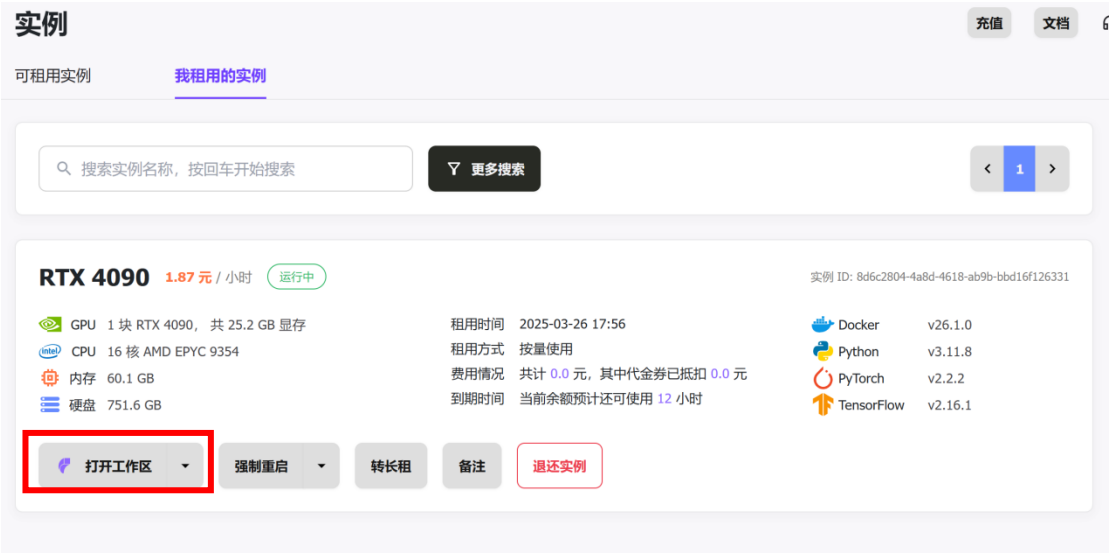
充值完成后，再点击左侧实例

选一块可用的显卡，点击开始使用

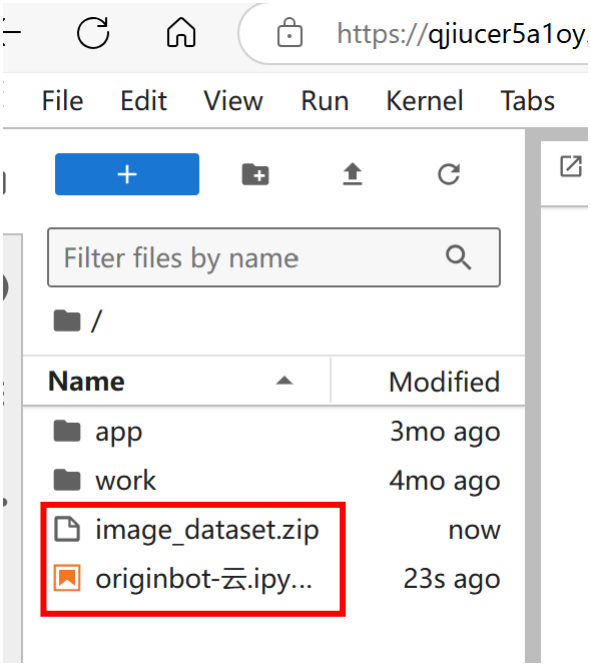

featurize
实例
数据集
笔记本
实例
充值
文档
可租用实例
我租用的实例
搜索实例名称，按回车开始搜索
更多搜索
RTX 4090
1 张可用
按量使用¥1.87/ 小时
GPU RTX 4090，共 25.2 GB 显存
CPU 16 核 AMD EPYC 9354
内存 60.1 GB
硬盘 751.6 GB
开始使用
RTX 4090
1 张可用
按量使用¥1.87/ 小时
GPU RTX 4090，共 25.2 GB 显存
CPU 16 核 AMD EPYC 9354
内存 60.1 GB
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RTX 4090
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CPU 16 核 AMD EPYC 9354
内存 60.1 GB
硬盘 751.6 GB
开始使用
RTX 4090
1 张可用
按量使用¥1.87/ 小时
GPU RTX 4090，共 24.0 GB 显存
CPU 14 核 AMD EPYC 7453
内存 64.4 GB
硬盘 400.0 GB
RTX 4090
1 张可用
按量使用¥1.87/ 小时
GPU RTX 4090，共 25.2 GB 显存
CPU 16 核 AMD EPYC 9354
内存 60.1 GB
硬盘 751.6 GB
RTX 4090
1 张可用
按量使用¥1.87/ 小时
GPU RTX 4090，共 25.2 GB 显存
CPU 16 核 AMD EPYC 9354
内存 60.1 GB
硬盘 751.6 GB


RTX 4090 1 张可用
显存：24 GB
CPU：16 x AMD EPYC 9354
内存：56 GB
磁盘：700 GB
Featurize 平台严禁任何形式的挖矿行为，如若发现账号封禁且不退款！
计费方式
按量
¥ 1.87/小时
日租
¥ 42
周租
¥ 289
月租
¥ 1211
实例会实时计费，优先使用代金券抵扣费用，当账户没有余额时实例会自动退回，请保持余额充足。
建议在长期租用之前，先使用该方式验证实例环境是否可用，验证成功后再切换为长期租用即可。
选择镜像
App（推荐）
可通过内置应用市场安装各种应用、例如 ComfyUI，SD 等
Docker v26.1.0
Python v3.11.8
PyTorch v2.2.2
TensorFlow v2.16.1
PyTorch（不再维护）
版本较新的环境
Docker v20.10.10
Python v3.10.12
PyTorch v2.0.1
TensorFlow v2.13.0
1 x RTX 4090
取消
开始使用

点击打开工作区

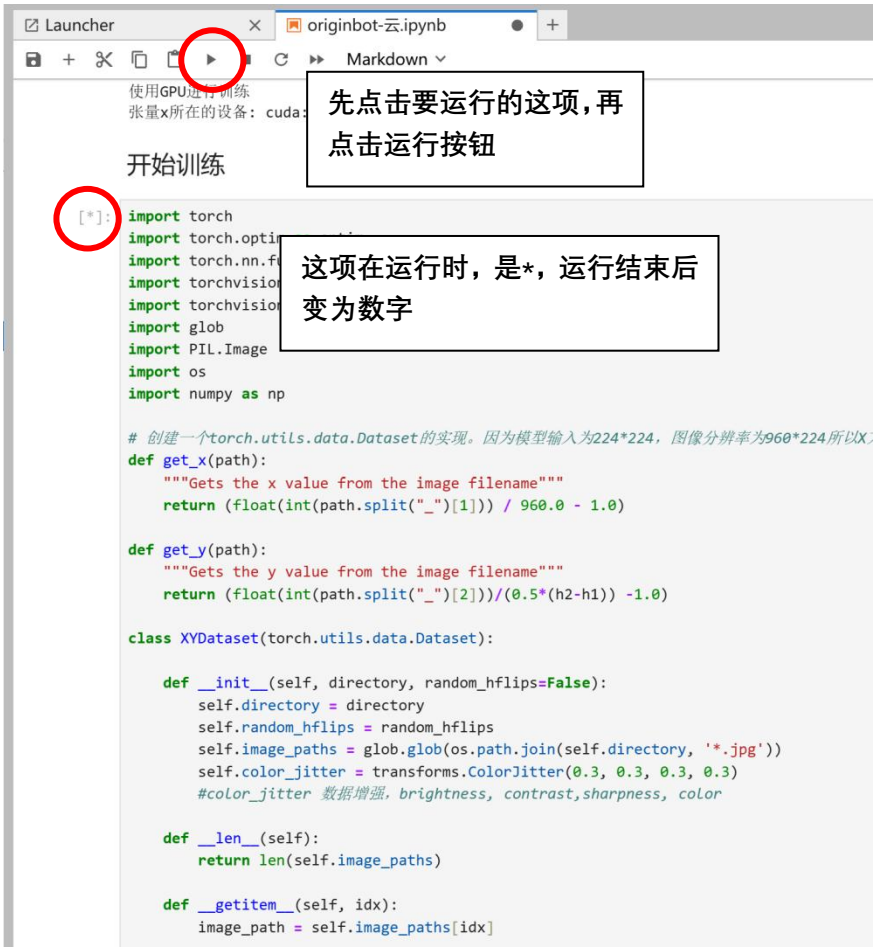


上传 image_dataset.zip 和 originbot-云-端.ipynb 上传到工作空间（用拖拉方式）



双击 originbot-云-端.jpynb

一项一项运行，要上一项运行结束后再点下一项运行。



把生成的 best_line_follower_model_xy.onnx 下载到计算机

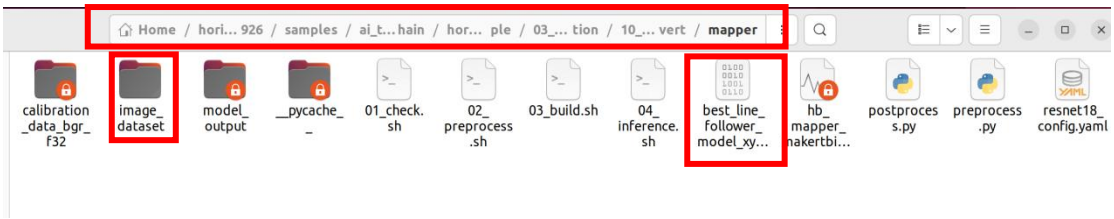
最后，返回，点击退还实例



3、模型转换（bin 格式）

打开虚拟机

①将标注好的数据集文件夹 image_dataset 和生成的 onnx 模型拷贝到以下 horizon_x5_open_explorer_v1.2.8-py310_20240926/samples/ai_toolchain/horizon_model_convert_sample/03_classification/10_model_convert/mapper/目录中



②*cd horizon_x5_open_explorer_v1.2.8-py310_20240926 /*
（用 table 补齐，不要直接 copy）
③*sudo bash run_docker.sh /data cpu*

```
originbot@originbot-virtual-machine:~/horizon_x5_c
0926$ sudo bash run_docker.sh /data cpu
[sudo] password for originbot:
root@dd1b4753c63e:/open_explorer#
```

密码：ros
④*cd samples/ai_toolchain/horizon_model_convert_sample/03_classification/10_model_convert/mapper/*
⑤*sh 02_preprocess.sh*

```
root@dd1b4753c63e:/open_explorer# cd samples/ai_toolchain/horizon_model_convert_
sample/03_classification/10_model_convert/mapper/
root@dd1b4753c63e:/open_explorer/samples/ai_toolchain/horizon_model_convert_samp
le/03_classification/10_model_convert/mapper# sh 02_preprocess.sh

cd $(dirname $0) || exit

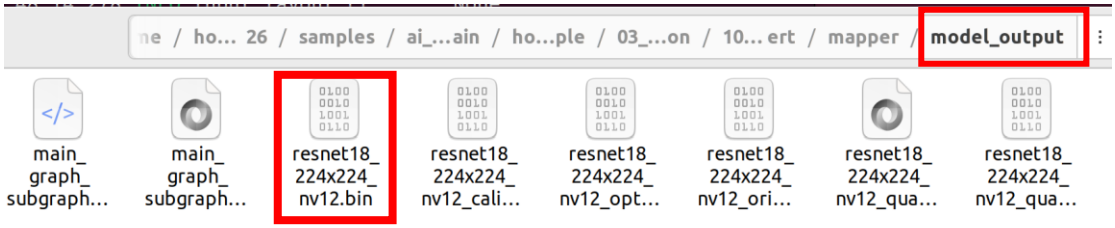
python3 ../../../data_preprocess.py \
  --src_dir /open_explorer/samples/ai_toolchain/horizon_model_convert_sample/03_
classification/10_model_convert/mapper/image_dataset \
  --dst_dir ./calibration_data_bgr_f32 \
  --pic_ext .rgb \
  --read_mode opencv
Warning please note that the data type is now determined by the name of the fold
er suffix
Warning if you need to set it explicitly, please configure the value of saved_da
ta_type in the preprocess shell script
Regular preprocess
Init 2 processes
write:./calibration_data_bgr_f32/xy_012_119_d583f7fe-cb50-11ef-936e-e4c767dc5914
.rgb
write:./calibration_data_bgr_f32/xy_010_133_d5e66eea-cb50-11ef-acbc-e4c767dc5914
.rgb
```

⑥*sh 03_build.sh*

```
le/03_classification/10_model_convert/mapper# sh 03_build.sh
2025-01-11 19:47:54,086 INFO log will be stored in /open_explorer/samples/ai_too
lchain/horizon_model_convert_sample/03_classification/10_model_convert/mapper/hb
_mapper_makertbin.log
2025-01-11 19:47:54,086 INFO Start hb_mapper....
2025-01-11 19:47:54,086 INFO hbdk version 3.49.15
2025-01-11 19:47:54,086 INFO horizon_nn version 1.1.0
2025-01-11 19:47:54,086 INFO hb_mapper version 1.24.3
2025-01-11 19:47:54,086 INFO Start Model Convert....
2025-01-11 19:47:54,091 INFO Using onnx model file: /open_explorer/samples/ai_to
olchain/horizon_model_convert_sample/03_classification/10_model_convert/mapper/b
est_line_follower_model_xy.onnx
2025-01-11 19:47:54,404 INFO Model has 1 inputs according to model file
2025-01-11 19:47:54,404 INFO Model name not given in yaml_file, using model name
 from model file: ['input']
```

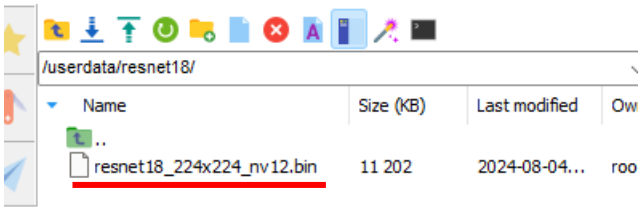
```
2025-01-11 19:48:14,278 INFO ---------input info : input ---------
2025-01-11 19:48:14,278 INFO input_name          : input
2025-01-11 19:48:14,278 INFO input_type_rt       : nv12
2025-01-11 19:48:14,278 INFO input_space&range   : regular
2025-01-11 19:48:14,278 INFO input_layout_rt     : None
2025-01-11 19:48:14,278 INFO input_type_train    : rgb
2025-01-11 19:48:14,279 INFO input_layout_train  : NCHW
2025-01-11 19:48:14,279 INFO norm_type           : data_mean_and_scale
2025-01-11 19:48:14,279 INFO input_shape         : 1x3x224x224
2025-01-11 19:48:14,279 INFO mean_value          : 123.675,116.28,103.53,
2025-01-11 19:48:14,279 INFO scale_value         : 0.0171248,0.017507,0.0174292,
2025-01-11 19:48:14,279 INFO cal_data_dir        : /open_explorer/samples/ai_too
lchain/horizon_model_convert_sample/03_classification/10_model_convert/mapper/ca
libration_data_bgr_f32
2025-01-11 19:48:14,279 INFO ---------input info : input end -------
2025-01-11 19:48:14,279 INFO ----------------------------------------
2025-01-11 19:48:14,279 INFO ############## calibration_parameters info #########
####
2025-01-11 19:48:14,279 INFO preprocess_on       : False
2025-01-11 19:48:14,279 INFO calibration_type:   : kl
2025-01-11 19:48:14,279 INFO per_channel         : False
2025-01-11 19:48:14,279 INFO ############## compiler_parameters info ############
#
2025-01-11 19:48:14,279 INFO debug               : False
2025-01-11 19:48:14,279 INFO compile_mode        : latency
2025-01-11 19:48:14,279 INFO hbdk_pass_through_params: --O3 --core-num 1 --fast
2025-01-11 19:48:14,279 INFO input-source        : {'input': 'pyramid', '_defaul
t_value': 'ddr'}
2025-01-11 19:48:14,293 INFO Convert to runtime bin file successfully!
2025-01-11 19:48:14,294 INFO End Model Convert
```

⑦拷贝 model_output/resnet18_224X224_nv12.bin 到当地计算上



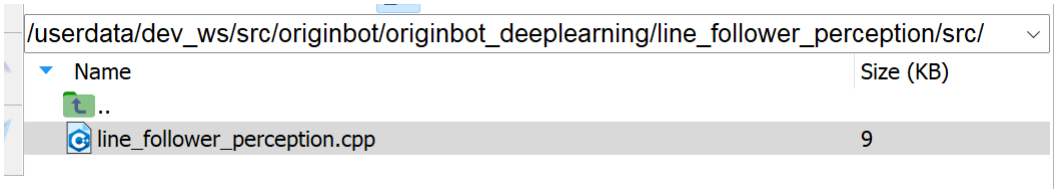
4、部署和推理

1）将文件 resnet18_224x224_nv12.bin 上传到/userdata/resnet18.



2）修改推理程序中的截图高度（h 值）

双击打开 /userdata/dev_ws/src/originbot/originbot_deeplearning/line_follower_perception/src/目录下的 line_follower_perception.cpp 程序。




```
189  hbDNNRoi roi;
190  roi.left = 0;
191  roi.top = 250;
192  roi.right = 1920;
193  roi.bottom = 474;
```

roi.top 的值改为 h1 值
roi.bottom 的值改为 h2 值

```
263  hbSysFlushMem(&(tensor.sysMem[0]), HB_SYS_MEM_CACHE_INVALIDATE);
264  float x = reinterpret_cast<float *>(te
265  float y = reinterpret_cast<float *>(te
266  result->x = (x +1.0) * 960.0;
267  result->y = 250+ (y + 1.0) *112.0;
268  RCLCPP_INFO(rclcpp::get_logger("LineFo
269              "coor rawx: %f,  rawy:%f,
270  return 0;
```

result->x= 反归一化 1.5 放大 10000 倍
result->y= 反归一化 0.2 放大 10000 倍

修改好了，存盘退出

3）推理程序进行编译

cd /userdata/dev_ws/

colcon build --packages-select line_follower_perception

```
root@ubuntu:~# cd /userdata/dev_ws/
root@ubuntu:/userdata/dev_ws# colcon build --packages-select line_follower_perception
Starting >>> line_follower_perception
Finished <<< line_follower_perception [0.91s]

Summary: 1 package finished [3.84s]
```

4）启动摄像头，输入

ros2 launch originbot_bringup camera.launch.py

```
root@ubuntu:/userdata/dev_ws# ros2 launch originbot_bringup camera_mipi.launch.py
[INFO] [launch]: All log files can be found below /root/.ros/log/2025-01-11-20-35-00-795692-ubuntu-4238
[INFO] [launch]: Default logging verbosity is set to INFO
config_file_path is  /opt/tros/humble/lib/mipi_cam/config/F37_calibration.yaml
Hobot shm pkg enables zero-copy with fastrtps profiles file: /opt/tros/humble/lib/hobot_shm/config/shm_fastdds.xml
Hobot shm pkg sets RMW_FASTRTPS_USE_QOS_FROM_XML: 1
env of RMW_FASTRTPS_USE_QOS_FROM_XML is  1 , ignore env setting
[INFO] [mipi_cam-1]: process started with pid [4239]
[mipi_cam-1] Wrong coordinate (-0.085153, 500.256134) (0.831019, 500.255859) (-0.085359, 501.244873) (0.830827, 501.244751), must bigger than 0
[mipi_cam-1] [ERROR] [1736598905.643402919] [mipi_cap]: hbn_gen_gdc_bin failed, ret = 0
[mipi_cam-1]
```

5）启动推理程序

SSH 连接小车，再开第 2 个窗口，输入

ros2 run line_follower_perception line_follower_perception

出现滚动，说明推理进行中。

```
[INFO] [1744607079.460216999] [LineFollowerPerceptionNode]: input size:1 roi size:1
[INFO] [1744607079.464448421] [LineFollowerPerceptionNode]: coor rawx: -0.145475,  rawy:0.786481, x: -0.218213    y:
0.117972
[INFO] [1744607079.477150726] [hobot_cv]: hobotcv_resize vps laps ms= 9
[INFO] [1744607079.477838644] [LineFollowerPerceptionNode]: input size:1 roi size:1
[INFO] [1744607079.481485356] [LineFollowerPerceptionNode]: coor rawx: -0.108079,  rawy:0.789660, x: -0.162118    y:
0.118449
[INFO] [1744607079.517939771] [hobot_cv]: hobotcv_resize vps laps ms= 10
[INFO] [1744607079.518690146] [LineFollowerPerceptionNode]: input size:1 roi size:1
[INFO] [1744607079.522081067] [LineFollowerPerceptionNode]: coor rawx: -0.103555,  rawy:0.791018, x: -0.155332    y:
0.118653
[INFO] [1744607079.545888884] [hobot_cv]: hobotcv_resize vps laps ms= 10
[INFO] [1744607079.546480802] [LineFollowerPerceptionNode]: input size:1 roi size:1
[INFO] [1744607079.549932347] [LineFollowerPerceptionNode]: coor rawx: -0.134715,  rawy:0.783154, x: -0.202072    y:
0.117473
[INFO] [1744607079.580972214] [hobot_cv]: hobotcv_resize vps laps ms= 7
[INFO] [1744607079.581540882] [LineFollowerPerceptionNode]: input size:1 roi size:1
[INFO] [1744607079.585106427] [LineFollowerPerceptionNode]: coor rawx: -0.147979,  rawy:0.794686, x: -0.221969    y:
0.119203
[INFO] [1744607079.621612925] [hobot_cv]: hobotcv_resize vps laps ms= 13
[INFO] [1744607079.622354468] [LineFollowerPerceptionNode]: input size:1 roi size:1
[INFO] [1744607079.626478764] [LineFollowerPerceptionNode]: coor rawx: -0.135530,  rawy:0.794754, x: -0.203295    y:
0.119213
[INFO] [1744607079.641749822] [hobot_cv]: hobotcv_resize vps laps ms= 10
[INFO] [1744607079.642431531] [LineFollowerPerceptionNode]: input size:1 roi size:1
[INFO] [1744607079.645968077] [LineFollowerPerceptionNode]: coor rawx: -0.123883,  rawy:0.786984, x: -0.185825    y:
0.118048
[INFO] [1744607079.687484747] [hobot_cv]: hobotcv_resize vps laps ms= 13
[INFO] [1744607079.688206039] [LineFollowerPerceptionNode]: input size:1 roi size:1
```

把小车放在车道线各个位置，观察 x 的值是否正确，此 x 表示转角。

6）在第三个终端，输入

ros2 topic echo /line_track_center_detection

```
---
header:
  stamp:
    sec: 1744607191
    nanosec: 884492478
  frame_id: map
point:
  x: 0.0
  y: 0.0
  z: 0.0
---
header:
  stamp:
    sec: 1744607191
    nanosec: 917826485
  frame_id: map
point:
  x: 0.0
  y: 0.0
  z: 0.0
---
header:
  stamp:
    sec: 1744607191
    nanosec: 950466074
  frame_id: map
point:
  x: 0.0
  y: 0.0
  z: 0.0
```